

6(a)(i)	<i>in context of mutations in genes</i> any two from: uncontrolled, (cell) division / mitosis ; cell cycle not regulated ; programmed cell death does not occur / cell becomes immortal ; uncontrolled growth not suppressed ; <i>ref. to</i> mass of abnormal cells ; AVP ; e.g. cell cycle checkpoints impaired proto-oncogenes become oncogenes tumour suppressor genes switched off	2
6(a)(ii)	any two from: <i>ref. to</i> change of a single, nucleotide / base (in sequence of DNA) or one base substitution (mutation) ; change in one, DNA triplet / mRNA codon ; <i>ref. to</i> tRNA with a different anticodon / tRNA brings a different amino acid ; idea that different primary structure leads to different tertiary structure ;	2

Question	Answer	Marks
6(b)(i)	<i>any three from:</i> vaccination stimulates an immune response ; antigen presentation, qualified ; e.g. TSA engulfed by, phagocyte / named phagocyte antigen binds to, specific / complementary, receptor, on, T-lymphocyte / cytotoxic T cell / killer T cell ; clonal expansion ; A described e.g. divides by mitosis to form a clone (receptor on) cytotoxic T cell / killer T cell, binds to TSA on tumour cell ; release of, perforin / hydrogen peroxide (to kill tumour cell) ; A other named e.g. granzymes AVP ; e.g. causes lysis of the cell	3
6(b)(ii)	<i>advantage any one relevant e.g.</i> tumour cells may have more than one TSA ; stimulate a specific immune response to each TSA ; <i>idea that</i> TSA on patient's tumour cells has the appropriate antigen <i>idea that</i> personalised treatment so more effective <i>disadvantage any one relevant e.g.</i> high cost of personalised treatment ; immune system may not recognise TSA on own body cell as, foreign / non-self , the cells in the tumour may not have the TSA in the vaccine ; patient's tumour may not be large enough to obtain enough cells for treatment ; patient needs surgery to obtain (large quantity of) cells ; AVP ; e.g. <i>suggestion that</i> tumour cell may settle in a part of the body and not be destroyed	2